

PART 1

```
SELECT c_firstname, c_lastname, o_orderkey, o_totalprice
FROM SNOWBEARAIR_DB.promo_catalog_sales.ORDERS
JOIN SNOWBEARAIR_DB.promo_catalog_sales.customer ON o_custkey = c_custkey
ORDER BY o_totalprice DESC
LIMIT 10;
```

```
INSERT INTO my_favorite_actors
VALUES
    ('Heath Ledger'),
    ('Michelle Pfeiffer'),
    ('Meryl Street'),
    ('Anthony Hopkins'),
    ('Bruce Lee');

SELECT * FROM my_favorite_actors;
```

The screenshot shows a SQL IDE interface. The top pane displays a SQL query with line numbers 5 through 17. The query includes a warehouse selection, a table listing command, and a select statement with a join. The bottom pane shows the results of the query, which is a table with 5 rows and 1 column named 'NAME'. The results are: Heath Ledger, Michelle Pfeiffer, Meryl Street, Anthony Hopkins, and Bruce Lee. On the right side of the bottom pane, there is a 'Query Details' panel showing a query duration of 267ms, 5 rows, and a query ID. Below this is an 'Ask Copilot' button.

```
5 USE WAREHOUSE CHEETAH_wh;
6
7 SHOW TABLES;
8
9 SELECT *
10 FROM members
11 LIMIT 10;
12
13 SELECT TOP 5
14     c.c_lastname,
15     c.c_firstname,
16     c.c_acctbal
17 FROM SNOWBEARAIR_DB.promo_catalog_sales.customer c
```

	NAME
1	Heath Ledger
2	Michelle Pfeiffer
3	Meryl Street
4	Anthony Hopkins
5	Bruce Lee

Query Details

Query duration 267ms

Rows 5

Query ID 01bba5a8-0304-f1f8-0...

Show more

NAME

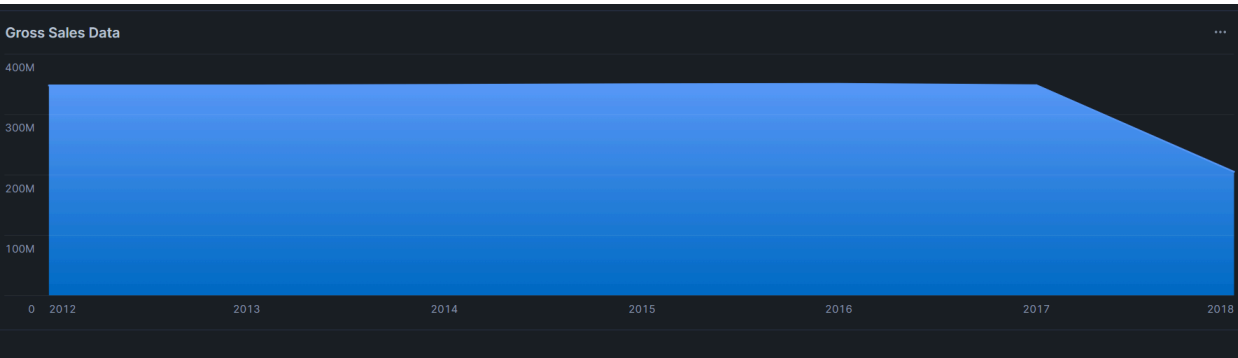
100% filled

Ask Copilot

PART 2

```
USE ROLE TRAINING_ROLE;
USE SCHEMA SNOWBEARAIR_DB.PROMO_CATALOG_SALES;
CREATE WAREHOUSE IF NOT EXISTS CHEETAH_wh;
USE WAREHOUSE CHEETAH_WH;

SELECT
  YEAR(O.O_ORDERDATE) AS year,
  SUM(L.L_EXTENDEDPRICE) AS sum_gross_revenue
FROM
  SNOWBEARAIR_DB.PROMO_CATALOG_SALES.CUSTOMER C
  INNER JOIN SNOWBEARAIR_DB.PROMO_CATALOG_SALES.ORDERS O ON C.C_CUSTKEY = O.O_CUSTKEY
  INNER JOIN SNOWBEARAIR_DB.PROMO_CATALOG_SALES.LINEITEM L ON O.O_ORDERKEY = L.L_ORDERKEY
GROUP BY year
ORDER BY year;
```



Gross Sales Data		7 rows	...
YEAR	SUM_GROSS_REVENUE		
2012	347,469,879		
2013	347,599,276		
2014	348,306,906		
2015	349,644,163		
2016	350,289,512		
2017	347,900,648		
2018	204,563,665		

PART 4

```
COPY INTO nation
FROM (
    SELECT
        n.$1 AS N_KEY ,
        n.$2 AS NATION ,
        CASE
            WHEN n.$3 = 1 THEN 'AMERICA'
            ELSE n.$3
        END AS R_KEY ,
        n.$4 AS COMMENTS
    FROM @training_db.traininglab.ed_stage/load/lab_files/nation.tbl (file_format => 'MYPIPEFORMAT') n
) FILE_FORMAT = (FORMAT_NAME = MYPIPEFORMAT) ON_ERROR = CONTINUE;
```

```
                                NATION VARCHAR ,
                                COMMENTS VARCHAR );

COPY INTO NATION
FROM @training_db.traininglab.ed_stage/load/lab_files/
FILES = ('nation.tbl')
FILE_FORMAT = (FORMAT_NAME = MYPIPEFORMAT)
VALIDATION_MODE = RETURN_ERRORS;
```

```
COPY INTO NATION
FROM @training_db.traininglab.ed_stage/load/lab_files/
FILES = ('nation.tbl')
FILE_FORMAT = (FORMAT_NAME = MYPIPEFORMAT)
VALIDATION_MODE = RETURN_10_ROWS;
```

```
CREATE OR REPLACE FILE FORMAT MYPIPEFORMAT
TYPE = CSV
COMPRESSION = NONE
FIELD_DELIMITER = '|'
FILE_EXTENSION = '.tbl' ERROR_ON_COLUMN_COUNT_MISMATCH = FALSE;

CREATE OR REPLACE FILE FORMAT MYGZIPPIPEFORMAT TYPE = CSV
COMPRESSION = GZIP
FIELD_DELIMITER = '|'
FILE_EXTENSION = '.tbl' ERROR_ON_COLUMN_COUNT_MISMATCH = FALSE;

CREATE OR REPLACE FILE FORMAT MYGZIPPIPEFORMAT TYPE = CSV COMPRESSION = GZIP FIELD_DELIMITER = '|' FILE_B

SELECT n.$1 FROM @training_db.traininglab.ed_stage/load/lab_files/region.tbl n;
```

PART 5

```
n.$2 AS NATION ,
CASE
WHEN n.$3 = 1 THEN 'AMERICA'
ELSE n.$3 END AS R_KEY ,
n.$4 AS COMMENTS FROM @training-db.traininglab.ed_stage/load

COPY INTO nation
FROM (
    SELECT
        n.$1 AS N_KEY ,
        n.$2 AS NATION ,
        CASE
            WHEN n.$3 = 1 THEN 'AMERICA'
            ELSE n.$3
        END AS R_KEY ,
        n.$4 AS COMMENTS
```

```
SELECT n.$1 AS N_KEY ,
CASE
WHEN n.$3 = 0 THEN 'AFRICA'
WHEN n.$3 = 1 THEN 'AMERICA'
WHEN n.$3 = 2 THEN 'ASIA'
WHEN n.$3 = 3 THEN 'EUROPE'
ELSE 'MIDDLE EAST'
END AS REGION ,
substr(REGION, 1, 2) AS REGION_CODE ,
n.$2 AS NATION ,
n.$4 AS COMMENTS
FROM @training-db.traininglab.ed_stage/load/lab_files/nation.tbl (file_format => 'MYPIPEFORMAT') n ;
```

Load Structured Data

part 6part 7

TRAINING_ROLECHEETAH_WH (X-Small)Share

CHEETAH_DB.WEATHERSettingsCode Versions

```
168 SELECT n.$1 AS N_KEY ,
169 CASE
170 WHEN n.$3 = 0 THEN 'AFRICA'
171 WHEN n.$3 = 1 THEN 'AMERICA'
172 WHEN n.$3 = 2 THEN 'ASIA'
173 WHEN n.$3 = 3 THEN 'EUROPE'
174 ELSE 'MIDDLE EAST'
175 END AS REGION ,
176 substr(REGION, 1, 2) AS REGION_CODE ,
177 n.$2 AS NATION ,
178 n.$4 AS COMMENTS
179 FROM @training-db.traininglab.ed_stage/load/lab_files/nation.tbl (file_format => 'MYPIPEFORMAT') n ;
180
181
182
183 COPY INTO nation FROM (
```

ResultsChart

	N_KEY	REGION	REGION_CODE	NATION	COMMENTS
1	0	AFRICA	AF	ALGERIA	haggle, carefully final deposits detect slyly agai
2	1	AMERICA	AM	ARGENTINA	ai foxes promise slyly according to the regular accounts, bold r
3	2	AMERICA	AM	BRAZIL	y alongside of the pending deposits, carefully special package
4	3	AMERICA	AM	CANADA	eas hang ironic, silent packages, slyly regular packages are fur
5	4	MIDDLE EAST	MI	EGYPT	y above the carefully unusual theodolites, final dugouts are qui
6	5	AFRICA	AF	ETHIOPIA	ven packages wake quickly, regu

Query Details

Query duration171ms

Rows25

Query ID01bba5dc-0304-f1fa-00...

Show more

Ask Copilot

PART 6

```

"pressure": 10161, "pressure-quality-code": "1" }, "dt": "2019-07-30T02:00:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 80, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 41 } }, { "air": { "dew-point": 8.2, "dew-point-quality-code": "1", "temp": 29.9, "temp-quality-code": "1" }, "atmospheric": { "pressure": 10126, "pressure-quality-code": "1" }, "dt": "2019-07-30T02:30:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 80, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 41 } }, { "air": { "dew-point": 32.2, "dew-point-quality-code": "1", "temp": 32.2, "temp-quality-code": "1" }, "atmospheric": { "pressure": 10126, "pressure-quality-code": "1" }, "dt": "2019-07-30T05:00:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 60, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 57 } }, { "air": { "dew-point": 6.4, "dew-point-quality-code": "1", "temp": 32.2, "temp-quality-code": "1" }, "atmospheric": { "pressure": 10123, "pressure-quality-code": "1" }, "dt": "2019-07-30T05:30:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 60, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 57 } }, { "air": { "dew-point": 30.6, "dew-point-quality-code": "1", "temp": 30.6, "temp-quality-code": "1" }, "atmospheric": { "pressure": 10123, "pressure-quality-code": "1" }, "dt": "2019-07-30T08:00:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 140, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 26 } }, { "air": { "dew-point": 5.1, "dew-point-quality-code": "1", "temp": 30.7, "temp-quality-code": "1" }, "atmospheric": { "pressure": 10152, "pressure-quality-code": "1" }, "dt": "2019-07-30T08:30:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 120, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 10 } }, { "air": { "dew-point": 18.9, "dew-point-quality-code": "2", "temp": 18.9, "temp-quality-code": "2" }, "atmospheric": { "pressure": 10152, "pressure-quality-code": "1" }, "dt": "2019-07-30T11:30:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 150, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 10 } }, { "air": { "dew-point": 3.5, "dew-point-quality-code": "1", "temp": 17.9, "temp-quality-code": "1" }, "atmospheric": { "pressure": 10165, "pressure-quality-code": "1" }, "dt": "2019-07-30T14:00:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 150, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 10 } }, { "air": { "dew-point": 5.5, "dew-point-quality-code": "1", "temp": 18, "temp-quality-code": "1" }, "atmospheric": { "pressure": 10165, "pressure-quality-code": "1" }, "dt": "2019-07-30T14:30:00", "sky": { "ceiling": 99999, "ceiling-quality-code": "9" }, "visibility": { "distance": 999999, "distance-quality-code": "9" }, "wind": { "direction-angle": 150, "direction-quality-code": "1", "speed-quality-code": "1", "speed-rate": 10 } }

```

```

CREATE OR REPLACE TABLE customers
AS
SELECT
    $1 AS id,
    parse_json($2) AS info
FROM VALUES
    (
        12712555,
        '{"name": {"first": "John", "last": "Smith"}, "contact": [ {"business": {"phone": "303-555-1234", "email": "j.smith@company.com"}, {"business": {"phone": "21-8322", "email": "jsmith332@gmail.com"}} ] }, {'
    ),
    ( 98127771,
        '{"name": {"first": "Jane", "last": "Doe"}, "contact": [ {"business": {"phone": "303-638-4887", "email": "jg_doe@company.com"}, {"business": {"phone": "789", "email": "happyjane@gmail.com"}} ] }, {'
    );

SELECT * FROM customers;

```

```

USE ROLE training_role;

CREATE WAREHOUSE IF NOT EXISTS CHEETAH_WH;
ALTER WAREHOUSE CHEETAH_WH SET WAREHOUSE_SIZE=XSmall;
USE WAREHOUSE CHEETAH_WH;

CREATE DATABASE IF NOT EXISTS CHEETAH_DB;
USE DATABASE CHEETAH_DB;

CREATE SCHEMA IF NOT EXISTS CHEETAH_SCHEMA;
USE SCHEMA CHEETAH_SCHEMA;

CREATE OR REPLACE TABLE customers AS
SELECT
    $1 AS id,
    parse_json($2) AS info
FROM VALUES
    (12712555, '{"name": {"first": "John", "last": "Smith"}}'),
    (98127771, '{"name": {"first": "Jane", "last": "Doe"}}');

SELECT * FROM customers;

```

```

SELECT query_id, substr(query_text, 1, 50) partial_query_text, user_name,
warehouse_name, warehouse_size,
BYTES_SPILLED_TO_REMOTE_STORAGE, start_time, end_time, total_elapsed_time/1000000
FROM snowflake.account_usage.query_history
WHERE BYTES_SPILLED_TO_REMOTE_STORAGE > 0 AND start_time::date > dateadd('day', -30, current_date)
ORDER BY BYTES_SPILLED_TO_REMOTE_STORAGE
DESC LIMIT 10;

```

```

SHOW WAREHOUSES; SELECT "name" AS WAREHOUSE_NAME ,
'size' AS WAREHOUSE_SIZE
FROM TABLE(RESULT_SCAN(LAST_QUERY_ID()))
WHERE "resource_monitor" = 'null';

WITH WAREHOUSE_SIZE AS ( SELECT WAREHOUSE_SIZE, "CREDITS/HOUR"
FROM ( SELECT 'XSMALL' AS WAREHOUSE_SIZE, 1 AS "CREDITS/HOUR"
UNION ALL SELECT 'SMALL' AS WAREHOUSE_SIZE, 2 AS "CREDITS/HOUR"
UNION ALL SELECT 'MEDIUM' AS WAREHOUSE_SIZE, 4 AS "CREDITS/HOUR"
UNION ALL SELECT 'LARGE' AS WAREHOUSE_SIZE, 8 AS "CREDITS/HOUR"
UNION ALL SELECT 'XLARGE' AS WAREHOUSE_SIZE, 16 AS "CREDITS/HOUR"
UNION ALL SELECT '2XLARGE' AS WAREHOUSE_SIZE, 32 AS "CREDITS/HOUR"
UNION ALL SELECT '3XLARGE' AS WAREHOUSE_SIZE, 64 AS "CREDITS/HOUR"
UNION ALL SELECT '4XLARGE' AS WAREHOUSE_SIZE, 128 AS "CREDITS/HOUR" ) ),
QUERY_HISTORY AS ( SELECT QH.QUERY_ID ,
QH.QUERY_TEXT ,
QH.USER_NAME ,
QH.ROLE_NAME ,
QH.EXECUTION_TIME ,
QH.WAREHOUSE_SIZE
FROM
SNOWFLAKE.ACCOUNT_USAGE.QUERY_HISTORY QH
WHERE START_TIME > DATEADD(month,-2,CURRENT_TIMESTAMP()) )

```

SNOWFLAKE.ACCOUNT_USAGE ▾ Settings ▾

```

USE ROLE TRAINING_ROLE;
CREATE WAREHOUSE IF NOT EXISTS CHEETAH_WH
INITIALLY_SUSPENDED=TRUE;
USE WAREHOUSE CHEETAH_WH;
USE SCHEMA SNOWFLAKE.ACCOUNT_USAGE;

-- Credits used (all time = past year)
SELECT WAREHOUSE_NAME ,
SUM(CREDITS_USED_COMPUTE) AS CREDITS_USED_COMPUTE_SUM
FROM ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY
GROUP BY 1
ORDER BY 2 DESC;

-- Credits used (past N days/weeks/months)
SELECT WAREHOUSE_NAME, SUM(CREDITS_USED_COMPUTE) AS CREDITS_USED_COMPUTE_SUM
FROM ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY
WHERE START_TIME >= DATEADD(DAY, -7, CURRENT_TIMESTAMP()) // Past 7 days
GROUP BY 1 ORDER BY 2 DESC;

```

